

UI Developer Interview Questions 2025

HTML & Browser Fundamentals

1. What's the difference between render tree, DOM tree, and CSSOM tree?
2. Explain the Critical Rendering Path and how to optimize it.
3. Why are `<script defer>` and `<script async>` used? What's the difference?
4. What is Content Security Policy (CSP) and why is it important?
5. Explain Shadow DOM and use cases.
6. What are microtasks vs macrotasks in the event loop?
7. How does layout thrashing happen and how to avoid it?

CSS & Modern Styling

1. Compare Tailwind CSS vs CSS-in-JS vs SCSS — what to use when?
2. How does the browser handle CSS specificity? Explain with examples.
3. What is container queries and how is it different from media queries?
4. Explain CSS Subgrid.
5. What is Cascade Layers (`@layer`)?
6. Difference between `display: block`, `inline`, `inline-block` and position properties.
7. Explain repaints and reflows.

JavaScript (Modern ES2025)

1. Event Loop, Call Stack, Task Queue, Microtask Queue.
2. Promise vs `async/await` vs Generators.
3. Memoization use cases.
4. Debounce vs throttle.
5. WeakMap and WeakSet uses.
6. Shallow copy vs deep copy.
7. Modules and import maps.
8. `requestIdleCallback` vs `requestAnimationFrame`.
9. Temporal API updates.

React (Latest)

1. Server components vs client components.

2. Suspense boundaries and streaming server rendering.

3. useTransition() usage.

4. React Fiber working.

5. memo, useMemo, useCallback.

Angular (2025)

1. Signals vs RxJS.

2. Deferrable views.

3. Zone-less Angular.

4. Standalone Components usage.

Performance Optimization

1. CLS, LCP, FID improvements.

2. Image optimization importance.

3. Fixing render-blocking resources.

4. Lazy loading, code splitting, preload, prefetch.

5. SSR vs CSR vs SSG vs ISR.

Accessibility (A11y)

1. WCAG 2.2 updates.

2. ARIA roles & live regions.

3. Keyboard accessibility.

4. Focus trapping.

5. Screen reader interpretations.

System Design for UI Developers

1. Designing a component library.

2. Micro-frontends.

3. Infinite scroll design.

4. Real-time UI updates.

5. Scaling UI for millions.

API + State Management

1. Redux vs Zustand vs Jotai.
2. Server State vs Client State.
3. TanStack Query internal working.
4. REST vs GraphQL vs tRPC.
5. Optimistic UI updates.

Testing

1. Unit vs integration vs E2E.
2. Playwright vs Cypress vs Jest vs RTL.
3. Mock Service Worker (MSW).

Security

1. XSS, Clickjacking, CSRF, CORS.
2. SameSite cookies.
3. OAuth 2.1 and PKCE.